Daily Drilling Report
SPD19B-03
3/21/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
11.0	24"	17.00	72.83	31.70	95.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
26" CP, 178.00mKB	0.00	213.60	606,913.73		9.10
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
875.00	875.00		0.00		

Operations at Report Time

Operations Summary

Establish circulation, Circulate 1.2 casing string volume. Perform CMT job. Prepare to skid to well # 05.

Operations next 24hrs

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammad Ali Rezaei	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	3/9/2014	3/31/2014
BOP Test		
Chemical Spill Drill		
Fire Drill	3/9/2014	3/11/2014
H2S Drill	3/16/2014	3/31/2014
Heli Crash Drill	2/20/2014	3/14/2014
Medical Drill		
MOB Drill	3/1/2014	3/23/2014
Pit Drill		
Trip Drill		

STOP Cards

Number of Reports:

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	02:00	2.00	24"	G	07	P	Cont' shaking up & down the string while keeping 1000psi inside CSG, establish circulation.
02:00	02:30	0.50	24"	G	18	P	Land CHH again on landing ring, Ok.
02:30	04:00	1.50	24"	G	07	P	M/U Swage and circulate 1.2 casing string volume w/ 450 gpm & 200psi.
04:00	05:00	1.00	24"	H	01	P	B/O circulation swage. M/U cementing X/O (18 5/8" BTC x Quick Union). Drop bottom plug. M/U Mehran Single cement head. (Top plug already loaded). Line up surface line to cement unit.
05:00	07:30	2.50	24"	H	09	P	HPJSM. Flush and pressure test surface line to 500 / 2000 psi for 5/10 min, Ok. Mix & pump 400 bbls of lead CMT slurry 12.5 ppg w/ 7-8 bpm, 330 - 460 psi, (Solution Volume: 311bbl). Cont' pump 210 bbls of tail cement slurry 15.8 ppg w/ 6.5 bpm, 290 - 380 psi, (Solution Volume: 124 bbl). Drop top cement plug , displace CMT slurry w/ 20 bbls of SW by CMT unit. * Totally CMT usage: 100 MT of CMT class "G") * Lead CMT usage: Bentonite: 3558lbs + Sodium Silicate: 151 gal + Retarder: 770 lbs + Defoamer: 50 gal + Dispersant: 850 lb. * Tail CMT usage: Retarder: 165 lbs + Defoamer: 43 gal + Dispersant: 800 lb.
07:30	09:30	2.00	24"	H	07	P	Switch to rig pump. Cont' displace CMT slurry with SW w/ 450 - 150 gpm, 280 - 850-340 psi, bump plug after 818 bbls (totally 838 bbls), final displacing pressure w/ 150 gpm & 340 psi. Increase pressure to 1000 psi & test CSG for 10 min, OK. Bleed off pressure and check for back flow, floats not holding. Work on floats by applying hydraulic shocks sevral times, finally holding.
09:30	10:00	0.50	24"	H	01	P	R/D Mehran 18 5/8" Single CMT head. B/O & L/D X/O.
10:00	11:00	1.00	24"	G	20	P	Rotate string 7 turns to right and back off CHH RT. B/O & L/D Landing Joint & RT.
11:00	12:00	1.00	24"	G	01	P	R/D AL-Ghaith 18 5/8" CSG handling equipment. Change 22 ft long bail to drilling bail. Prepare for Skid over well SPD19B-05. Off-line job: Top Job on wells SP19B-08 & 03 : L/U CMT unit to 26" x 18 5/8" annulus, # 08 filled up after pumping 13 bbls of 12.5 ppg CMT slurry, # 03 not filled up after pumping 135 bbls of 12.5 ppg CMT slurry.

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Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
pH	Pf (mL/mL)	Mf (mL/mL)	Pm (mL/mL)	Sand (%)	Low Gravity Solids (%)	High Gravity Solids (%)
Vis 3rpm	Vis 6rpm	Percent Oil (%)	Percent Water (%)	Solids (%)	Chlorides (mg/L)	Filter Cake (/32")
Gel 10 sec (lbf/100ft²)	Gel 10 min (lbf/100ft²)	Gel 30 min (lbf/100ft²)	Oil Water Ratio	T (fl) (°C)	Active Mud Volume (bbl)	Reserve Mud Volume (bbl)
Cum Mud Added (bbl)	Mud Lost to Surface (bbl)	Mud Lost to Hole (bbl)	Avg Background Gas (%)	Avg Drill Gas (%)	K	n

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
C-Mariner	Supply Boat			
DP3	Supply Boat			
Osam Condor	Supply Boat			

Mud Additives

Description	Consumed	Cum Rec	On Loc
Bentonite Bulk	0.0	139.0	96.5
CALCIUM CHLORIDE DH	17.0	80.0	63.0
Lime	0.0	195.0	96.0
Caustic Soda	2.0	427.0	233.0
Soda Ash	0.0	193.0	84.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Drill Water	0.0	100.0	433.0
Diesel Fuel	7.0	0.0	145.0
Potable Water	22.0	16.0	96.0
Bentonite Bulk	0.0	0.0	88.5
Bentonite BB	0.0	0.0	22.0
Cement G	120.0	70.0	133.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	153.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
TFA (in²)	Drilling Time (hrs)	Bit Revs	Depth In (mKB)	Depth Out (mKB)	Depth Drilled (m)	BHA ROP (m/hr)	IADC Bit Dull
Flow Rate (min) (gpm)	Flow Rate (max) (gpm)	SPP (min) (psi)	SPP (max) (psi)	WOB (min) (1000lbf)	WOB (max) (1000lbf)	RPM (min) (rpm)	RPM (max) (rpm)
Circulating Time (hrs)	Rotating Time (hrs)	% Rotating Time (%)	% Sliding Time (%)	PU HL (max) (1000lbf)	Rot HL (max) (1000lbf)	SO HL (min) (1000lbf)	String Wt (1000lbf)

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
Applied Surface Pressure (psi)	Volume Pumped (bbl)	Current MAASP	Kick Tolerance	Dens Fluid (lb/gal)	Leak Off Pressure (psi)

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
Asmari		374.50	372.44
Jahrum		493.00	487.29
Ilam		847.00	

General Remarks

Comment